Sean Luka Girgis

Senior Data Engineer | AWS, PySpark, SQL & Data Platform Modernization

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Professional Summary

Senior Data Engineer with 20+ years of enterprise IT experience, including banking-scale data platforms, Python/PySpark ETL, SQL optimization, AWS data services, and production observability. Strong background designing and supporting large-scale pipelines, governed data workflows, reporting layers, and operational controls in regulated financial environments. Currently expanding hands-on modern data stack capabilities across Snowflake, dbt, Delta Lake, Unity Catalog, and Terraform, with a practical bridge from AWS Glue, Athena, S3, Redshift, Spark, SQL, and enterprise governance.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Architected automated Python/Pandas ETL pipelines ingesting P95 telemetry from 6,000+ enterprise endpoints, replacing manual processes with governed, repeatable data workflows.
- Built a hybrid AWS/Oracle data platform using S3 landing zones, AWS Glue ETL, Redshift analytics workloads, and Oracle retention schemas to support scalable forecasting and reporting.
- Designed optimized Oracle schemas and historical retention structures for large telemetry datasets, supporting seasonal analysis, performance review, and long-running enterprise reporting needs.
- Developed PySpark and Python processing patterns for large-scale telemetry and capacity datasets, improving throughput, maintainability, and repeatability across pipeline workloads.
- Created executive dashboards and reporting layers using Streamlit, matplotlib, seaborn, and plotly to turn infrastructure telemetry into stakeholder-ready business and operational insight.
- Integrated monitoring feeds from BMC TrueSight/TSCO, CA Wily, and AppDynamics into unified reporting datasets, improving traceability and consistency across data sources.
- Applied data quality, validation, and reconciliation practices to telemetry and reporting pipelines, supporting trusted outputs for capacity, provisioning, and infrastructure decision-making.
- Collaborated with engineering, operations, and leadership teams in a regulated banking environment, translating complex platform issues into clear findings, next steps, and production improvements.
- Supported production readiness through monitoring, alerting, incident analysis, runbook-style operational practices, and proactive issue prevention across data and observability workflows.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Managed Dynatrace AppMon and Gomez Synthetic Monitoring for Brand.com and other critical systems, supporting production observability, transaction tracing, and performance analysis.
- Led the FAST project, mining real-user and synthetic performance metrics to identify optimization opportunities adopted by engineering teams.
- Built operational analytics and QLIK reporting around application performance data, helping stakeholders understand customer-impacting trends and system behavior.
- Upgraded Dynatrace from 6.5 to 7.0 and enforced TLS 1.2 security with zero production monitoring regression during cutover.

- Integrated HP Performance Center with Dynatrace to correlate load testing, transaction behavior, and APM telemetry for release readiness.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Served as CA APM SME for TIAA-CREF, managing 50+ Enterprise Managers and 4,000–6,000 instrumented agents across a large financial-services environment.
- Designed dashboards, alerts, thresholds, and operational reporting modules adopted as standards across client environments.
- Built Perl and Korn Shell data-extraction automation for weekly performance reporting, reducing manual export effort and improving consistency.
- Led large-scale CA APM upgrades with zero unplanned downtime, coordinating engineering, operations, and application teams.
- Provided technical documentation, training, and troubleshooting guidance to improve support readiness and standardize operational practices.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed J2EE telecom applications under load to identify throughput limits, JDBC bottlenecks, thread constraints, heap pressure, CPU saturation, and garbage-collection issues.
- Installed and configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts for real-time visibility during performance testing.
- Documented performance baselines and regression thresholds used by engineering teams as release gate inputs.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster supporting transaction volumes at extreme scale.
- Optimized core transaction processing using C++ and OCCI, reducing physical hardware footprint by 95% while maintaining sub-second query latency.
- Built C++/OCCI/OCI validation frameworks to verify migration correctness across millions of transaction records.
- Automated schema conversion and data validation processes to protect integrity through phased cutover.

Architect/Developer (IRS CADE Project) at Computer Science Corporation (CSC) (2007-10 – 2008-05)

- Designed UML class and sequence diagrams and developed modernization modules for the IRS CADE project.
- Built CICS, MQ Series, XML, and VC++ interfaces backed by DB2 while following government architecture standards.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (AMDOCS projects) (2001 – 2007)

- Developed high-availability C++ billing interfaces using POSIX threads, mutex locks, semaphores, socket programming, and Oracle Pro*C/PLSQL patterns.
- Improved telecom billing processes by reducing memory usage 75%, improving throughput 20%, and achieving 10x database performance through sequence optimization.

- Maintained and enhanced AMDOCS billing, settlement, reporting, and interface modules across the full billing lifecycle.
- Automated WebLogic and WebSphere administration using Korn Shell scripting, reducing manual deployment and operational steps.

Education

High Diploma / Post-Graduate Diploma in Computer Engineering Technology / Computer Science
Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering
Zagazig University, Egypt / Cairo, Egypt

Skills

Flagship Projects

Serverless Lakehouse Platform (AWS)
Designed a serverless data platform using S3, Glue, Athena, Bedrock, and PySpark patterns.
Built Text-to-SQL GenAI capabilities and optimized Parquet/Snappy data access for scalable analytics and governed data consumption.
Technologies: AWS Glue, Athena, S3, PySpark, Bedrock, Parquet, SQL

HorizonScale — AI Capacity Forecasting Engine
Built an AI-driven capacity forecasting engine using Python, PySpark, Prophet, scikit-learn, and Streamlit.
Replaced manual forecasting workflows, created stakeholder-facing dashboards, and supported forecasting against banking-scale telemetry data.
Technologies: Python, PySpark, Prophet, scikit-learn, Streamlit, Data Pipelines

AI-Powered Job Search Pipeline
Built an end-to-end agentic automation pipeline with semantic duplicate detection, LLM-powered scoring, tailored document generation, quality gates, and tracking.
Demonstrates practical Python orchestration, RAG-style retrieval, structured outputs, and AI-assisted development workflow design.
Technologies: Python, FAISS, Sentence Transformers, Grok API, Claude Code, RAG, Agentic AI Workflows